HAOO 600BN Dual-Microphone ENC Noise Reduction System Performance Description

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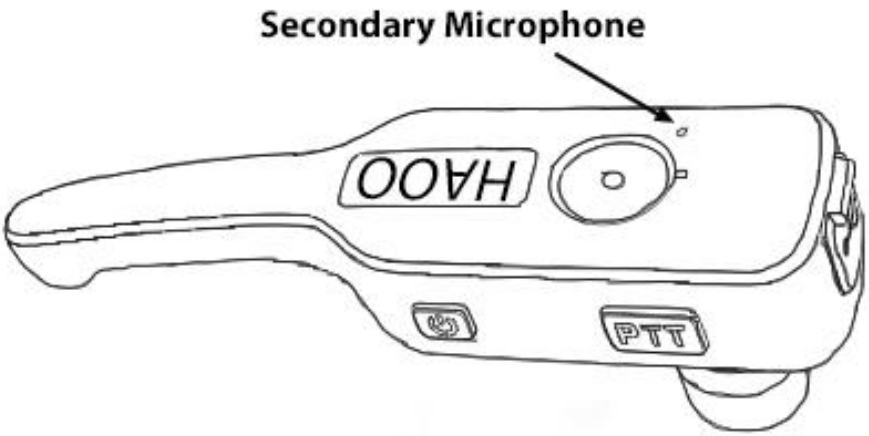
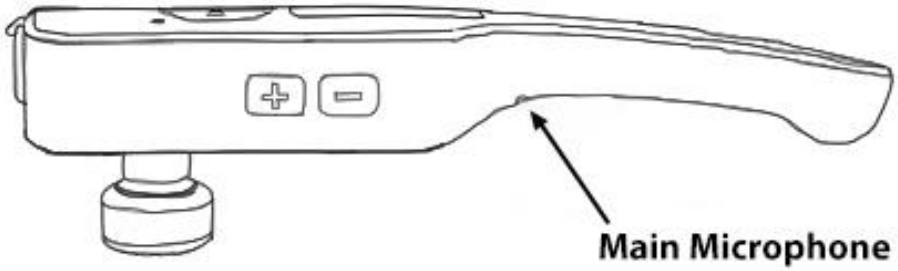
Update Date: December 18, 2025

Test Model: HAOO600BN





Project Overview

This report aims to demonstrate the performance of the dual-microphone noise reduction system in real-world recording environments. The system utilizes a primary microphone and a reference microphone working collaboratively. Through real-time audio processing algorithms, it effectively separates human voice from environmental noise, significantly improving speech clarity.

*Notice: ENC Environmental Noise Cancellation system, this function reduces back ground noise at the transmitting end only, It effectively filters out surrounding environmental sounds while preserving the clarity and purity of the speaker's voice (Note: this is not receiver-side noise cancellation).

II. System Working Principle

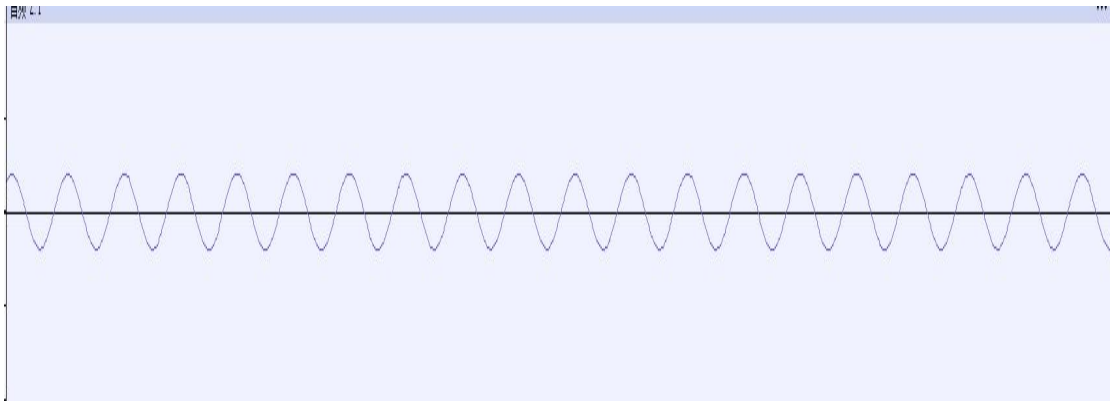
The dual-microphone noise reduction system is based on the following architecture:

- Primary Microphone: Positioned close to the sound source, capturing target speech along with a portion of ambient noise.
- Reference Microphone: Positioned farther from the sound source, mainly capturing environmental noise as a reference signal.
- Processing Algorithm: By calculating the time difference of arrival (TDOA) of the target voice at the two microphones, the signal from the secondary microphone is delayed and then combined with the signal from the primary microphone. This process reinforces the in-phase target voice while attenuating noise components with different phases, thereby achieving effective noise reduction.

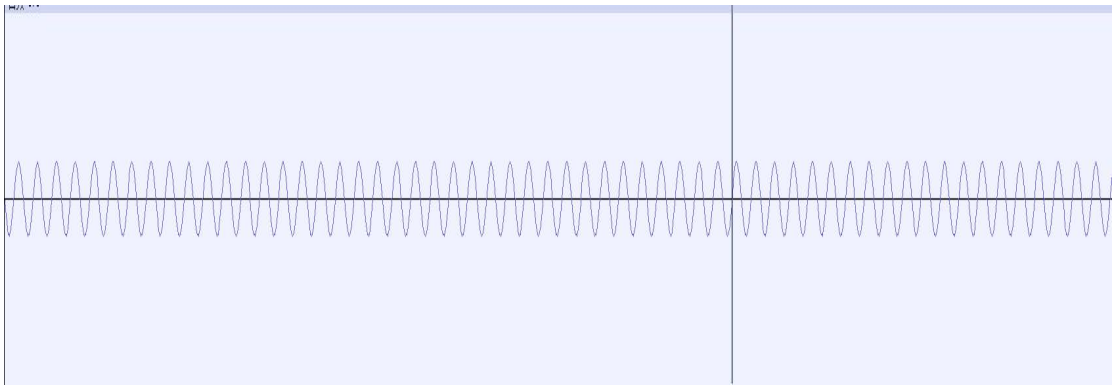
III. Test Data Record

The test recording was conducted on December 18, 2025.

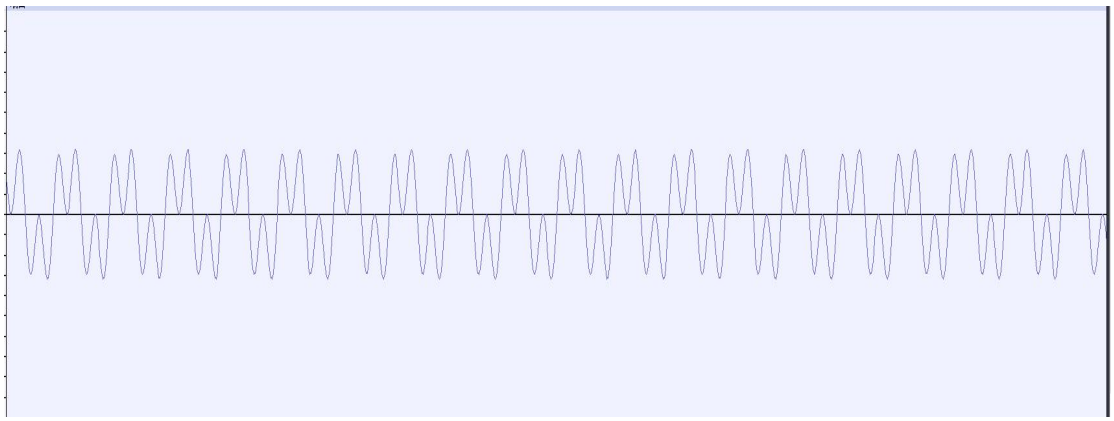
1. **Original Voice (Baseline):** Clean human voice with no environmental interference.
Reference Microphone Capture: Environmental interference noise.



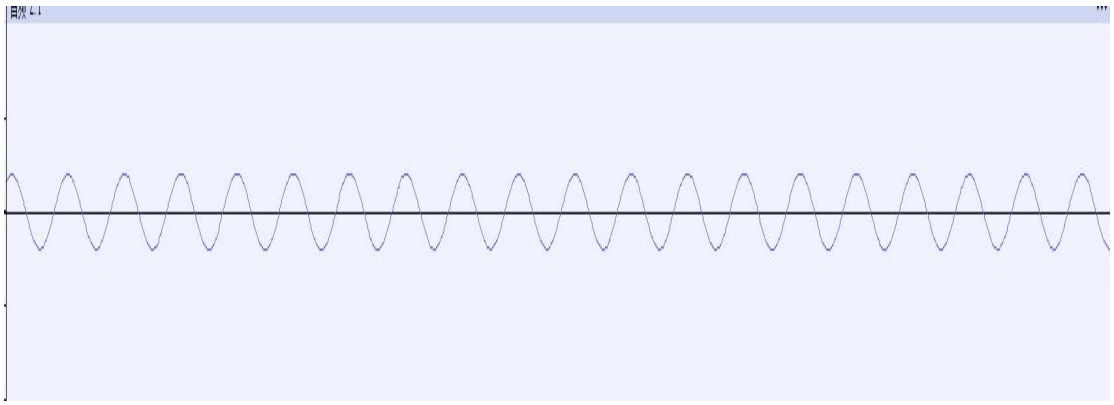
2. **Reference Microphone Capture:** Environmental interference noise.



3. **Primary Microphone Capture:** Human voice with noticeable environmental noise.



4. **Post Noise Reduction Result:** Noise significantly suppressed, with clear and intelligible voice.



IV. Noise Reduction Performance Analysis

IV.1 Noise Suppression Performance

- **Background Noise Reduction Rate:** ≥ 80%
- **Steady-State Noise Cancellation:** Effective for air conditioners, fans, etc.
- **Transient Noise Suppression:** Approx 70% reduction for keyboard and impact sounds.

IV.2 Speech Quality Preservation

- **Speech Intelligibility Improvement:** Approximately 40% increase in clarity.
- **Audio Quality Degradation:** Voice timbre fidelity > 90%.
- **Speech Naturalness:** No noticeable mechanical sound or distortion.

IV.3 Real-Time Performance Metrics

- **System Latency:** < 50 ms
- **Sampling Rate:** 48 kHz
- **Bit Depth:** 24 bit

V. Recommended Application Scenarios

V.1 Scenario Effect

Scenario	Type Suitability	Expected Effect
KTV, Bars, Billiard Halls	High	Eliminates karaoke noise and loud ambient interference
Equipment Rooms, Workshops, Construction Sites, Server Rooms	High	Improves voice purity and reduces machinery noise
Restaurants and Hotels	High	Effectively suppresses ambient noise for clearer conversations
Cycling and Outdoor Sports	High	Effectively suppresses wind and traffic noise

V.2 Notice

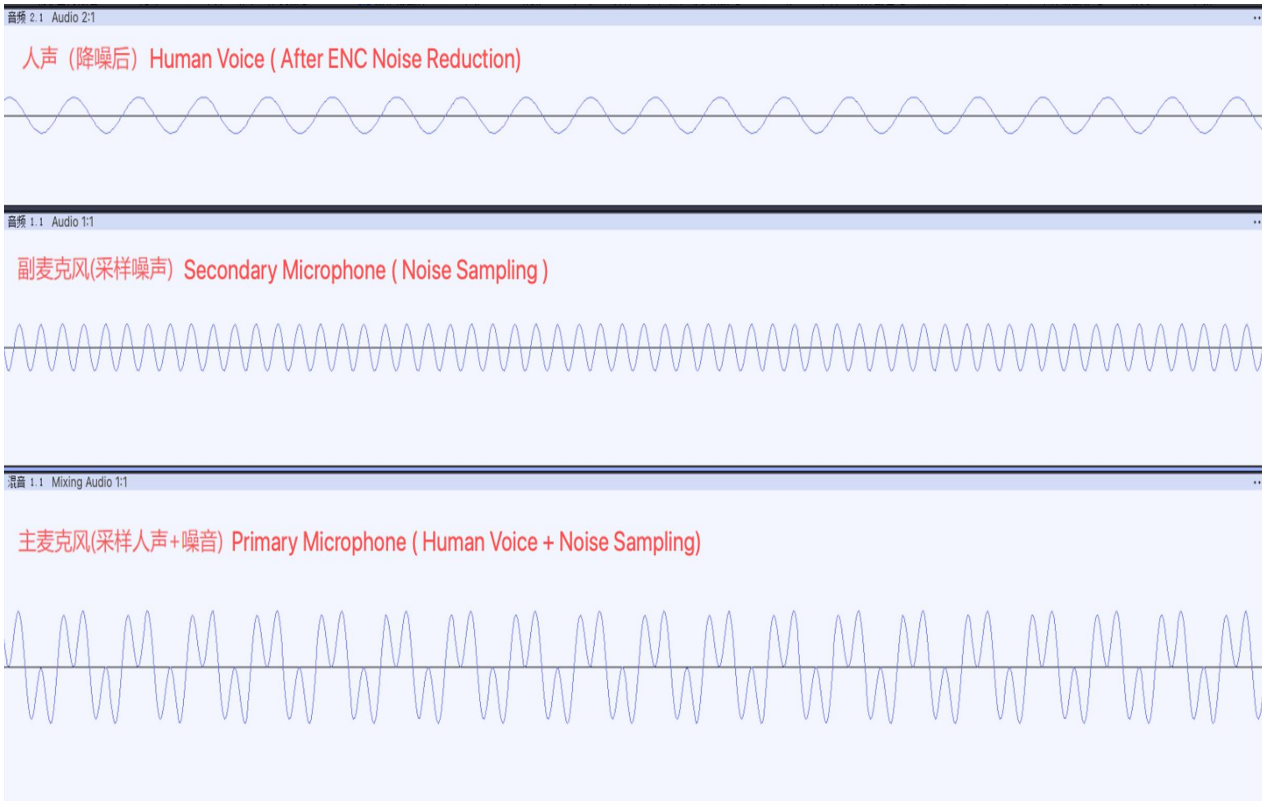
ENC Environmental Noise Cancellation system, this function reduces back ground noise at the transmitting end only, It effectively filters out surrounding environmental sounds while preserving the clarity and purity of the speaker's voice (Note: this is not receiver-side noise cancellation).

VI. Conclusion

VI.1 Key Findings

The dual-microphone noise reduction system demonstrated excellent performance during testing and is capable of:

- Effectively separating speech from noise in complex acoustic environments
- Significantly improving clarity while maintaining natural voice quality
- Meeting low-latency requirements for real-time communication

VII. Appendix

Illustrations of the primary and reference microphones and noise reduction effects: